MCQ A · domain and properties

After S47; MCQ B after S50.

For each question, only one option is correct. Just checking the solutions.

1. The domain of $\ln(2x-1)$ is:

A. $\mathbb{R}$ B. $[\frac{1}{2};+\infty[$

C. $]0;+\infty[$ D. $]\frac{1}{2};+\infty[$

2. For $a,b>0$, $\ln(ab)$ is:

A. $\ln a+\ln b$ B. $\ln a \ln b$

C. $\ln(a+b)$ D. $a\ln b$

3. $\ln(e^{-3})$ is:

A. 3 B. e^3

C. -3 D. $\ln 3$

4. The derivative of $\ln(1+x^2)$ is:

A. $\frac{1}{1+x^2}$ B. $\frac{2x}{1+x^2}$

C. $2\ln x$ D. $\frac{2}{x}$

5. The equation $\ln x = 2$ has solution:

A. $2e$ B. $\ln 2$

C. 2 D. e^2

MCQ B · threshold, growth and parameter

For each question, only one option is correct. Justify your choice orally before checking the solutions.

6. The smallest integer $n \geq 0$ such that $(\frac{1}{2})^n < 0.1$ is:

A. 3 B. 10

C. 4 D. 2

7. For $0 < q < 1$ and $0 < a < 1$, $q^n < a$ amounts to:

A. $n > \frac{\ln a}{\ln q}$ B. $n < \frac{\ln a}{\ln q}$

C. $n > \frac{\ln q}{\ln a}$ D. $n > a/q$

8. $\lim_{x\to+\infty} \frac{\ln x}{x}$ is:

A. 1 B. 0

C. $+\infty$ D. $-\infty$

9. $\ln(x^2) = 2\ln x$ is valid:

A. for every real x B. for every $x \neq 0$

C. only for $x > 1$ D. for every $x > 0$

10. The maximum of $g(x) = \frac{\ln x}{x}$, $x>0$, is:

A. 1 B. e

C. $1/e$ D. 0

True or false · justification expected. For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. $\ln(1+1) = \ln 1 + \ln 1$.

2. Dividing an inequality by ln 2 reverses its direction.

3. $\ln x / x = 1$ has a positive solution.